

Overarching:

1. THERE IS SO MUCH HERE! It's clear that you spent a lot of time on this :) But also this is more than we would expect from them for a single homework assignment.
2. I went through this myself and I can safely say that I don't expect any student to be able to do 1.7 or 2.3-2.5. I think that dropping those parts would make it a good length, but I'm not sure if that wrecks some of the things you were going for with this assignment?
3. A lot of these questions have multiple parts to them that should be broken apart. We've found that when we try to pack multiple parts into a single questions students will inevitably not do all of the parts.

Specific Comments:

- 1.1
 - So the coefficient to be fit is just 1?
 - One issue we may run into is that we usually tell the students that, if `curve_fit` returns 1, it usually means that it didn't work!
- 1.2:
 - Why do you ask them to only fit the central 16 data points? Is there a reason to choose a subset and not the entire dataset?
- 1.3:
 - They don't have a great grasp of noise. I think that most of them have some understanding of scatter in a plot, but calling it Gaussian Noise isn't something I would expect them to know.
 - I really love the idea of having them talk about whether their fit parameter matches the exact parameter!
- 1.4
 - Wait, why are we fitting a quintic? This might need some kind of transition. I think talking about how adding in more fit parameters means that we have more flexibility in our fit (maybe?).
 - I don't expect them to know what the term quintic means (or even 5th degree polynomial). So, at the very least, we're going to need to give them the expression for a quintic function.
- 1.5
 - What would be the exact fit parameters for a quintic being fit to noisy data? Is it the same thing as a linear function?
 - I'm still not sure why we're fitting a quintic function? Or why we are only using 16 data points?
 - Also, small thing, but it's noisy, not noizy.
- 1.6

- I would recommend having the function take the y data and model fit y-values, not the y-values, x-values, and the best fit coefficients. That way that you don't have to make a new function for each of the different fits!
 - If you do decide to keep the current function definition, it's not clear that there should be two functions (one for linear and one for quintic). (This is why it makes more sense for the function to take the y_values and y_fit values.)
- 1.7
 - Wait, is this even theoretically possible? Fitting the quintic to a subset of noisy data in such a way that the MSE will always be less than the linear fit?
 - I know that students will not be able to do this.
 - Going through your solution, I may be misinterpreting what the question is asking. I thought it was that you had to fit quintics to different subsamples of the noisy data, to find which one gave you an MSE smaller than a linear fit? And then confirm that that subsample worked by testing it with 100 different noisy samples? But that doesn't seem to be what your solution does. It looks like you just picked the range 3:18? Is there a reason why this would give a better fit?
 - I'm not sure what I should be taking away from this figure?
- 1.8
 - I would get rid of the third question. We usually try to push students to use all of their data when fitting. This is mostly because they struggle with the concept of an outlier, and usually just decide that any data that doesn't fit their model should be discarded (as opposed to potentially picking a better model).
- 2.2
 - I needed to use the describe function to figure out which dataset is weird. So I would definitely explicitly prompt them to use that!
 - I would also prompt them to make a plot of the data, so they can see what it looks like!
 - I would not trust students to have a good idea about what possible functions they can use here. I would recommend giving them a list, like in the day 15 assignment.
 - Is it supposed to be the case that a linear function will meet the requirements? Because my $y = a * x + b$ gave me an MSE of 0.099
 - Also, I think there's some typos here. It should be "particle," (which you have later on!) not "partical."
- 2.3
 - No student will know how to do this.
 - But also, distribute in what sense?

- It might be the hot sun beating down on me, but I did not follow what this question was asking me to do. I think I mirroring the data about a point of symmetry?
- 2.4
 - Again, this might be the sun beating down on me, but I did not think to use the mean here. I actually didn't have a good sense of how I would leverage the data symmetry. I'm sure that if I thought about it some more it would make sense. But yeah, I definitely don't expect students would get this.